MediMesh

Medicine Availability & Pharmacy Request System: Solution, Tech Stack & 24-Hour Plan

Team	Pixel Minds
Problem	Problem Statement 38: Medicine Availability & Pharmacy Request System
Domain	Software & Web Development
Format	24-hour hackathon

Pitch line: Hum dawai dhundhne ko nahi, dawai milne ko solve karte hain.

1. Problem ka saar

Customer ko apni preferred pharmacy mein dawai nahi milti, aur pharmacies ke paas limited stock hota hai jise incoming requests ke beech manage karna padta hai. Judges ka checklist:

- **Must have:** search, availability, reservation, stock validation, status tracking
- **Stretch:** nearby pharmacy ranking, low-stock alerts
- **Edge cases:** quantity stock se zyada, dawai kisi pharmacy mein nahi, multiple customers limited stock maangein
- **Demo:** limited stock wali dawai search, competing requests, stock-aware reservation

Insight: zyada tar teams basic search + reserve CRUD app banayengi. Jeetne ke liye edge cases ko hi hero banao.

2. Teen versions ka comparison

	V1: Reservation Engine	V2: Liquidity Network	V3: Enhanced
Core idea	Soft-lock, partial fulfilment, fair queue	Request pharmacies ko broadcast, peer-to-peer borrow, ledger	V2 + scored matching, expiry routing, OCR, outbreak, emergency mode
Statement se match	Sabse seedha	Seedha + twist	Bahut door ja sakta hai
Build effort	Easy (6-8 hr)	Medium	Hard, 24 hr mein poora nahi hoga
Wow factor	Medium	High	Bahut high, adhura dikha to ulta
Startup potential	Low, features copy ho jaate hain	High, network effect	High, par focus bikhra
Risk	Baaki teams jaisa lagega	Logistics ka sawaal	Over-scope, demo toot sakta hai

Seekh: V1 safe hai par jeet nahi dilata. V3 impressive hai par sab kuch banane wali team aksar kuch bhi poora nahi dikha paati. V2 ka core sabse strong hai aur V1 ke best parts uske andar fit ho jaate hain. V3 se sirf wohi lo jo demo mein dikh sake.

3. Teeno versions ki detail

V1: Stock-aware Reservation Engine

- **Soft-lock reservation:** reserve karte hi stock held (BookMyShow seat lock jaisa), 30 min expiry. Confirm to deduct, reject/timeout to stock wapas. Race condition DB transaction/row lock se solve.
- **Smart partial fulfilment:** 10 chahiye, 6 hain: 6 yahan + 4 nearby pharmacy se, ya baaki restock pe notify.
- **Substitute suggestion:** same salt wali generic alternative. Sirf suggestion, final decision pharmacist/doctor ka, UI mein disclaimer.
- **Fair queue:** FIFO ya urgency flag (prescription, chronic patient) ke basis pe priority.
- **Stretch:** nearby ranking (distance + availability + response speed), low-stock aur back-in-stock alerts.

V2: Medicine Liquidity Network

Customer pharmacy ko nahi dhundta, request khud pharmacies ko dhundhti hai. Request kabhi fail nahi hoti.

- **Broadcast, not search:** request nearby 5-10 pharmacies ko jaati hai, jo pehle confirm kare + ETA de use mil jaata hai.
- **Peer-to-peer borrowing:** preferred pharmacy A ke paas stock nahi to B se borrow, customer A ke counter se hi leta hai. A ka customer bacha, B ko extra sale.
- **Inter-pharmacy ledger:** har borrow ka auto-settlement at wholesale + small fee.
- **Lost Sales Meter:** failed/partial requests ko rupees mein convert, pharmacist ko paying-customer banane wala feature.
- **Demand signal:** area mein achanak demand badhe to nearby pharmacies ko pre-stock alert.

V3: Enhanced (6 layers)

- **Scored matching + cascading fallback:** score = distance + stock confidence + response speed + fulfilment history. Top 3 ko pehle, timeout pe agla batch.
- **Expiry-aware liquidity:** near-expiry stock flag aur borrow requests mein pehle route, discount ke saath (dead-stock recovery).
- **Trust & safety:** Schedule H dawai ke liye prescription + pharmacist verification, licence-verified badge, substitute pharmacist approval ke baad, audit trail.
- **Prescription OCR + multi-medicine cart:** ek photo se poori prescription ka ek saath sourcing.
- **Outbreak mode:** 'Simulate dengue outbreak' button, demand heatmap garam, pre-stock alerts.
- **Impact modes:** emergency (insulin/antidote priority), chronic refill reminders, generic nudge.

4. Final solution: MediMesh

Chemist shops ka virtual warehouse. Dawai milne ki guarantee.

Layer	Kya aata hai	Kahan se
Foundation	Stock soft-lock, partial/split fulfilment, fair queue (saare edge cases cover)	V1
Hero	Auto-borrow jab preferred pharmacy ka stock zero, ledger entry, Ready at A	V2
Booster 1	Scored matching + cascading fallback (tech depth)	V3
Booster 2	Lost Sales Meter + expiry-aware routing (paisa + finance angle)	V2 + V3
Booster 3	Outbreak simulation button (demo showstopper)	V3
Safety	Prescription-only verification, substitute sirf suggestion, disclaimers	V3
Slide/fake	OCR, emergency mode, chronic refill, runner (ek 'runner assigned' status kaafi)	V3

Request lifecycle (status tracking)

Requested -> **Sourcing** (broadcast / borrow) -> **Confirmed** (pharmacy accept, stock locked) -> **Ready at counter** -> **Completed**. Side states: **Partial offer**, **Rejected**, **Expired** (lock timeout, stock released).

Edge cases kaise handle hote hain

Edge case	MediMesh ka jawab
Quantity stock se zyada	Split sourcing: A se 6 + B se 4, ya partial + restock notify
Dawai kisi pharmacy mein nahi	Substitute (same composition) suggestion + back-in-stock alert
Multiple customers, limited stock	Row-level soft-lock, baaki ko partial/alternative/queue, pharmacy dashboard pe 3 requests, stock for 2

5. Tech stack

Layer	Choice	Kyun
Frontend	Next.js (React) + Tailwind CSS	Fast UI, Vercel pe 1-click deploy
Backend/DB	Supabase (Postgres)	Free tier, SQL transactions, row locks
Realtime	Supabase Realtime	Do browser windows mein live status update
Map	Leaflet + OpenStreetMap	Free, API key nahi chahiye, animated polyline possible
Logic	Postgres functions (RPC) / Next.js API routes	Soft-lock aur borrow atomic transaction mein
Expiry job	Supabase pg_cron ya setInterval simulation	Lock timeout pe stock release
Deploy	Vercel + backup screen recording	Venue wifi ka risk
Seed data	10 pharmacies, 40 medicines, 3-4 low-stock	Demo scenario controlled rahe

Database schema (simple)

```
pharmacies(id, name, lat, lng, response_score, verified)
medicines(id, name, salt, price, rx_required)
inventory(pharmacy_id, medicine_id, qty_total, qty_locked, expiry_date)
requests(id, customer, medicine_id, qty, preferred_pharmacy_id,
         status, created_at)
reservations(id, request_id, pharmacy_id, qty, lock_expires_at, status)
ledger(id, from_pharmacy, to_pharmacy, medicine_id, qty, amount, ts)
```

```
demand_events(medicine_id, area, ts)   -- outbreak/heatmap
```

Soft-lock logic (idea)

```
-- atomic lock: only succeeds if enough free stock
UPDATE inventory
   SET qty_locked = qty_locked + :qty
 WHERE pharmacy_id = :p AND medicine_id = :m
   AND (qty_total - qty_locked) >= :qty;
-- rows_affected = 0 -> stock nahi, borrow/partial flow chalao
-- confirm: qty_total -= qty, qty_locked -= qty
-- reject/expire: qty_locked -= qty
```

Matching score (V3 booster)

score = w1 x (1/distance) + w2 x stock_confidence + w3 x response_speed + w4 x fulfilment_history.
Top 3 pharmacies ko pehle request, 60-90 sec timeout ke baad agla batch (cascading fallback).

6. 24-hour execution plan

Time	Kaam	Output
0-2 hr	Repo, Next.js + Tailwind, Supabase, tables, seed data (3-4 medicines ka stock 2-5)	App chal raha, data DB mein
2-6 hr	Customer flow: search, availability list, request submit, stock validation (partial/split option)	Search -> request kaam kare
6-10 hr	Soft-lock reservation (transaction, 30 min expiry), status tracking, pharmacy dashboard, realtime	Do windows mein live status
10-15 hr	Hero: auto-borrow, nearby lock, ledger, 'Ready at A', cascading fallback	Borrow flow end-to-end
15-19 hr	Leaflet map + animated stock-flow line, Lost Sales Meter, outbreak button (heatmap + alerts)	Wow moments ready
19-21 hr	Bug fixing, edge case testing, safety disclaimers UI	Demo bina toote chale
21-23 hr	Slides, demo script ki 3-4 rehearsals, Q&A prep	Pitch ready
23-24 hr	Freeze + rehearsal, koi naya feature nahi. Vercel deploy + backup recording	Submit

Team split (3-4 log)

- **Backend/DB:** schema, soft-lock transaction, borrow logic, ledger
- **Frontend:** customer UI + pharmacy dashboard
- **Map/demo features:** Leaflet, animation, outbreak button, Lost Sales Meter
- **Pitch:** slides, business model, Q&A (frontend ko bhi support kare)

Akele ya 2 log ho to outbreak button aur scored matching kaat do, baaki sab rakho.

Golden rules

- Pehle 15 ghante mein demo ka core chalna chahiye, baaki polish hai.
- Jo dikhna nahi hai use hardcode/slide pe rakho (OCR, runner, emergency mode).
- 'Reset demo' button banao jo stock initial state pe la de.
- Demo ki screen recording backup rakho.
- Kam se kam 3-4 ghante ki neend, thake dimaag se demo bigadta hai.

7. 3-minute demo flow

1. Limited-stock medicine search (stock sirf 5).
2. Do customers ek saath 4-4 maangein: ek lock ho, doosre ko partial + alternative offer mile.
3. Preferred pharmacy ka stock zero: map pe borrow animation, ledger entry, 'Ready at A in 12 min'.
4. Pharmacy dashboard pe Lost Sales Meter: network ke bina itne rupees ki sale jaati.
5. Outbreak button: heatmap + pre-stock alerts.
6. Closing: Hum dawai dhundhne ko nahi, dawai milne ko solve karte hain.

8. Startup story

- **Revenue:** 3-5% transaction fee per borrow + pharmacy SaaS dashboard (inventory + request management, monthly subscription).
- **Moat:** network effect, zyada pharmacies = kam failed requests = zyada data.
- **Data product:** demand heatmap (kaunsi dawai baar-baar out of stock) distributors aur pharma companies ke liye.

- **Vision:** ek sheher ki saari chemist shops ek virtual warehouse. Aage: chronic refill, delivery partner integration, dead-stock recovery commission.

9. Risks aur Q&A tayyari

Sawal	Jawab
Logistics (runner) real mein kaise?	Demo mein simulated status. Real mein local delivery partners ya pharmacy-to-pharmacy runner, pilot ek locality se.
Regulation / drug laws?	Schedule H ke liye prescription + pharmacist verification, licence-verified pharmacies, audit trail. Deployment se pehle legal review.
Substitute galat de diya to?	Sirf suggestion, pharmacist/doctor approval ke baad hi, disclaimer UI mein.
Pharmacy kyun judegi?	Lost Sales Meter rupees dikhata hai, dead-stock bikta hai, extra sales milti hain.
Competition (1mg, PharmEasy)?	Wo delivery/e-commerce hain. Hum local chemist ke beech inventory liquidity network hain.

Judging fit: Innovation (P2P stock network + expiry liquidity) | Technical depth (locks, scored matching, realtime) | Feasibility (safety layer + simulated logistics) | Business (fees + SaaS) | Impact (emergency, refill, generic nudge).